GROCERY STORE SALES ANALYSIS

Power BI Student Project Report

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Project Title	Grocery Store Sales Analysis using Microsoft Power BI
Project Scope	Items, Categories, Customers and Sales
Tools Used	Microsoft Power BI Desktop, Microsoft Excel and DAX

Project Overview

This project uses Power BI to analyse grocery store data in four defined areas: **Items, Categories, Customers and Sales**. It includes a data model, DAX measures, dashboard visuals and key observations. The workbook used for this project contains synthetic practice data, not actual transactions from a real store.

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1. Dataset Description

The workbook contains four related tables: 10,000 sales transactions dated in 2025, 80 items, 10 categories and 1,000 customers. The data is designed for practising reporting, relationships and aggregations in Power BI.

Table	Columns	Purpose
Items	ItemID, ItemName, CategoryID, UnitPrice	Product details and unit price.
Categories	CategoryID, CategoryName	Groups items into grocery categories.
Customers	CustomerID, CustomerName, City	Customer identity and city.
Sales	SaleID, Date, CustomerID, ItemID, Quantity, UnitPrice, SalesAmount	Transactions, units sold and sales value.

2. Project Objectives

• Analyse total sales, transaction count and quantity sold.

• Identify high-performing items based on sales value.

• Review categories and compare category-level sales.

• Identify customers contributing the highest sales.

• Visualise monthly sales trends.

• Build a clear report using Power BI visuals and DAX.

3. Tools and Technologies

Microsoft Excel: data storage. **Power BI Desktop:** data modelling and visualisation. **DAX:** measures and aggregations. The scope is limited to Items, Categories, Customers and Sales.

4. Data Preparation and Cleaning

The workbook uses consistent column names and separates product, category and customer information into lookup tables. In Power Query and Model view, verify data types, blanks/errors, duplicate IDs, valid relationships, and numeric types for Quantity, UnitPrice and SalesAmount. Date should use the Date type; SalesAmount should use Decimal Number or Fixed Decimal Number.

Reporting note: These are recommended validation checks. Only report that a particular error or duplicate was removed if you actually verified and performed that step.

5. Data Model

Use one consistent set of tables to avoid duplicate or ambiguous relationship paths. The intended model uses CategoriesTable filtering ItemsTable, then Sales; CustomersTable filters Sales. Relationships should be one-to-many, active and single-direction.

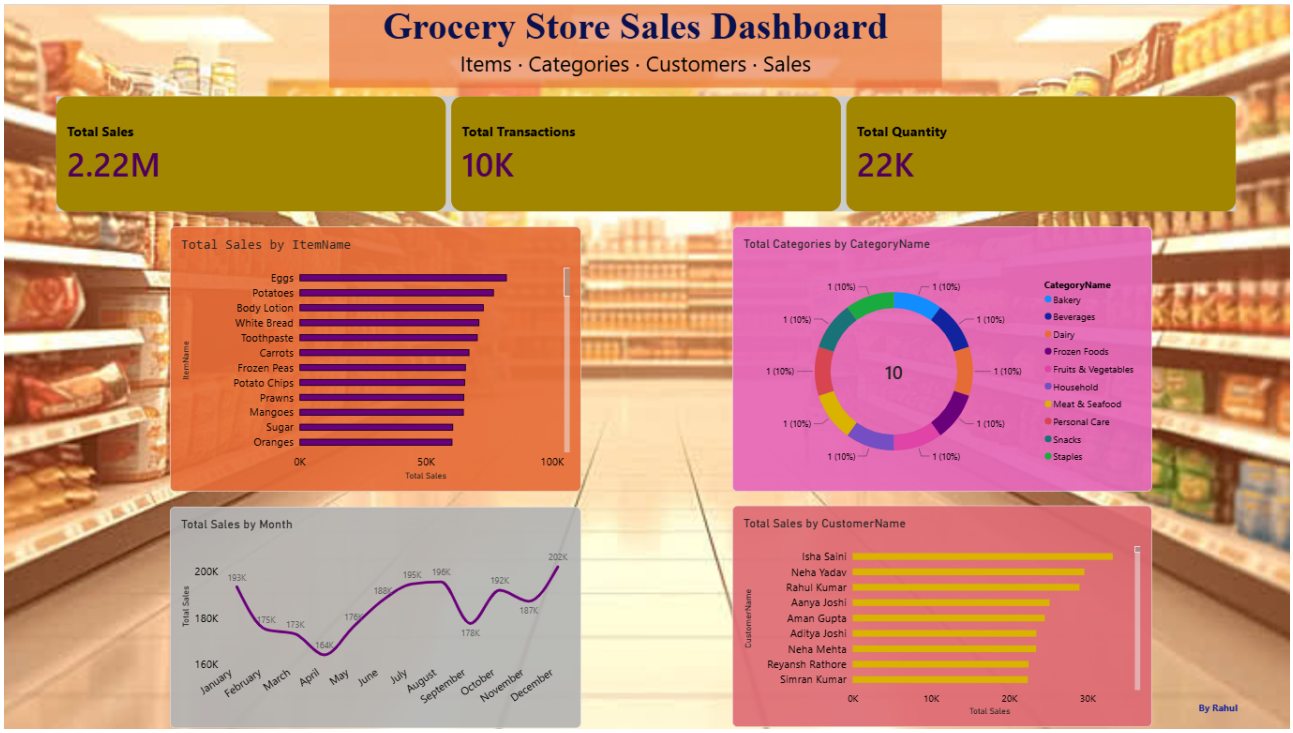
Lookup table	Relationship	Fact table column
CategoriesTable	CategoryID (1) → CategoryID (*) in ItemsTable	Filters Sales through ItemsTable
ItemsTable	ItemID (1) → ItemID (*)	Sales[ItemID]
CustomersTable	CustomerID (1) → CustomerID (*)	Sales[CustomerID]

6. DAX Measures

Measure	DAX formula
Total Sales	`Total Sales = SUM(Sales[SalesAmount])`
Total Transactions	`Total Transactions = DISTINCTCOUNT(Sales[SaleID])`
Total Quantity	`Total Quantity = SUM(Sales[Quantity])`
Total Customers	`Total Customers = DISTINCTCOUNT(CustomersTable[CustomerID])`
Total Items	`Total Items = DISTINCTCOUNT(ItemsTable[ItemID])`
Total Categories	`Total Categories = DISTINCTCOUNT(CategoriesTable[CategoryID])`

7. Dashboard Design and Visuals

The current report page combines KPI cards and charts for total sales, transactions, quantity, item sales, category count, monthly sales trend and customer sales. The screenshot below documents the dashboard layout.



Visual Mapping

Visual	Fields	Purpose
KPI cards	Total Sales, Total Transactions, Total Quantity	Summarise overall performance.
Item bar chart	ItemsTable[ItemName] + [Total Sales]	Compare items by sales value.
Category donut	CategoryName + [Total Categories] in current screenshot	Shows equal category counts; use [Total Sales] in Values for category sales comparison.
Monthly line chart	Sales[Date] + [Total Sales]	Show monthly sales trend.
Customer bar chart	CustomerName + [Total Sales]	Compare customer sales contribution.

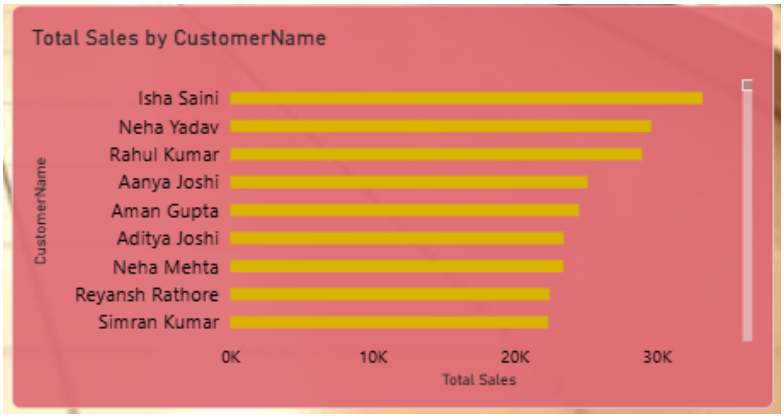
8. Key Findings

The dashboard screenshot displays these summary values. They describe the synthetic practice dataset and should not be interpreted as audited business results.

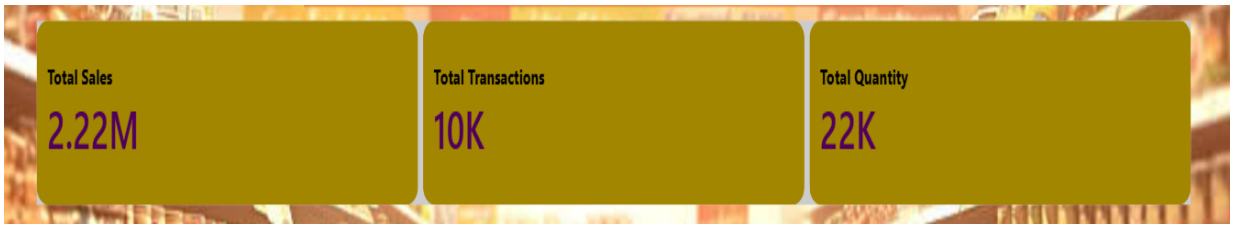
Metric	Displayed result	Interpretation
Total Sales	2.22M	Aggregate sales value across the transaction records.
Total Transactions	10K	Approximately 10,000 transaction records.
Total Quantity	22K	Approximately 22,000 units sold.
Total Categories	10	Ten grocery categories are present.
Monthly trend	April ~164K; December ~202K	Visible monthly sales fluctuate and finish higher in December.

9. Screenshot Evidence

Customer sales chart



KPI cards



10. Conclusion

The Grocery Store Sales Analysis project demonstrates how Power BI can convert structured transaction data into a visual report. The model separates item, category and customer information from the Sales transaction table. DAX measures summarise sales, transactions, quantity, items, categories and customers. Bar charts support item and customer comparisons, while the line chart shows the monthly sales trend.

Before final submission, update the category donut chart to use Total Sales as its Values field if the objective iscategory-wise sales analysis. Also verify that only one consistent set of lookup tables and the Sales fact table remains activein the model.

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